CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 12: Algebraic Identities

Topic-wise short notes with solved examples

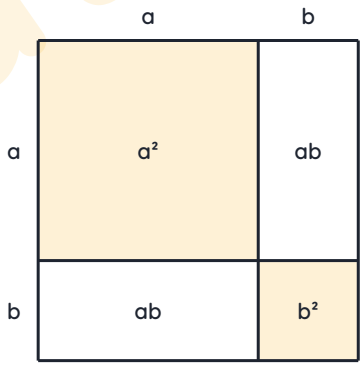
The standard identities for the square of a sum and a difference, the product of a sum and a difference, the square of a trinomial and the cubes of binomials, with applications.

1 Identity and equation

- An **identity** is an equality that is true for **all** values of its variables. $(x+2)^2 = x^2 + 4x + 4$ is an identity.
- An **equation** is true only for certain values. $x + 2 = 5$ is true only when $x = 3$.

2 The basic identities

Identity	Expansion
$(a+b)^2$	$a^2 + 2ab + b^2$
$(a-b)^2$	$a^2 - 2ab + b^2$
$(a+b)(a-b)$	$a^2 - b^2$
$(x+a)(x+b)$	$x^2 + (a+b)x + ab$



A square of side a + b has area a² + 2ab + b²

Example 2.1 Expand $(3x+5y)^2$ and $(4a-3b)^2$.

SOLUTION

$9x^2 + 30xy + 25y^2$ **and** $16a^2 - 24ab + 9b^2$

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Example 2.2 Expand $\left(2x + \frac{1}{2x}\right)^2$.

SOLUTION

$4x^2 + 2(2x)\left(\frac{1}{2x}\right) + \frac{1}{4x^2}$

$4x^2 + 2 + \frac{1}{4x^2}$

Example 2.3 Find the product $(5m + 2n)(5m - 2n)$.

SOLUTION

$25m^2 - 4n^2$

Example 2.4 Find the product $(x + 7)(x - 3)$.

SOLUTION

$x^2 + (7 - 3)x + (7)(-3)$

$x^2 + 4x - 21$

3 Quick calculation

- Write the number as a sum or difference involving a round number.

Example 3.1 Evaluate 103^2, 97^2 and 105×95.

SOLUTION

$103^2 = 10000 + 600 + 9$; $97^2 = 10000 - 600 + 9$; $105 \times 95 = 100^2 - 5^2$

10609, 9409 and 9975

Example 3.2 Evaluate $(8.3)^2 - (1.7)^2$.

SOLUTION

$(8.3 + 1.7)(8.3 - 1.7) = 10 \times 6.6$

66

Example 3.3 Evaluate 52×53.

SOLUTION

$(50 + 2)(50 + 3) = 2500 + 250 + 6$

2756

4 Relations between the identities

- $(a+b)^2+(a-b)^2=2(a^2+b^2)$
- $(a+b)^2-(a-b)^2=4ab$
- $a^2+b^2=(a+b)^2-2ab=(a-b)^2+2ab$
- $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2=\left(x-\frac{1}{x}\right)^2+2$

Example 4.1 If $a+b=8$ and $ab=15$, find a^2+b^2.

SOLUTION

$64-30$

34

Example 4.2 If $a-b=5$ and $ab=14$, find a^2+b^2 and $(a+b)^2$.

SOLUTION

$a^2+b^2=25+28=53;\ (a+b)^2=25+56$

53 and 81

Example 4.3 If $x+\frac{1}{x}=6$, find $x^2+\frac{1}{x^2}$.

SOLUTION

$36-2$

34

Example 4.4 If $x-\frac{1}{x}=3$, find $x^2+\frac{1}{x^2}$ and $x^4+\frac{1}{x^4}$.

SOLUTION

$9+2=11$; then 11^2-2

11 and 119

5 Square of a trinomial

$$(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$$

- For a negative term, carry its sign into the products.

Example 5.1 Expand $(x+2y+3z)^2$.

SOLUTION

$x^2+4y^2+9z^2+4xy+12yz+6zx$

Example 5.2 Expand $(2a - b + c)^2$.

SOLUTION

$4a^2 + b^2 + c^2 - 4ab - 2bc + 4ca$

Example 5.3 If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$.

SOLUTION

$81 - 52$

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6 Cubes of binomials

- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$
- $x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

Example 6.1 Expand $(2x + 3)^3$.

SOLUTION

$8x^3 + 3(4x^2)(3) + 3(2x)(9) + 27$

$8x^3 + 36x^2 + 54x + 27$

Example 6.2 Expand $(a - 2b)^3$.

SOLUTION

$a^3 - 6a^2b + 12ab^2 - 8b^3$

Example 6.3 If $a + b = 4$ and $ab = 3$, find $a^3 + b^3$.

SOLUTION

$a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 64 - 36$

28

Example 6.4 If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.

SOLUTION

$27 - 9$

18

Example 6.5 Evaluate $(99)^3$.

SOLUTION

$(100-1)^3 = 1000000 - 30000 + 300 - 1$

970299

7 Common mistakes

- Writing $(a+b)^2 = a^2 + b^2$ and leaving out $2ab$.
- Writing $(a-b)^2 = a^2 - b^2$.
- Taking the last term of $(a-b)^2$ as $-b^2$. It is $+b^2$.
- Forgetting to square the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.
- Using $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 + 2$. The sign is minus.
- Missing the factor 3 in the middle terms of a cube.

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